Design an app and a responsive website that helps parent and students find bus routes and stop information. the app should also let parents and students search for or create their own carpooling groups

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Project overview

The product:

This project is a mobile app and responsive website designed to help parents and students easily find school bus routes, stops, and schedules in one place. The platform also enables users to search for, join, or create trusted carpooling groups within their community, making daily school commutes safer, more organized, and collaborative. The primary target users are school students and their parents who need reliable, real-time transportation information and coordinated ride-sharing options.

Project duration:

January 2026 – March 2026

Project overview

The problem:

Parents and students struggle to access clear, centralized information about school bus routes and stop locations. Existing information is often scattered, outdated, or difficult to understand, and there is no reliable platform to coordinate safe and trusted carpooling within their community.

The goal:

Design a user-friendly mobile app and responsive website that provides accurate bus route and stop information while enabling parents and students to easily search, create, and manage trusted carpooling groups for safer and more organized school commutes.

Project overview

My role:

Lead UX Designer and UX Researcher responsible for designing the end-to-end experience for both the mobile app and responsive website.

Responsibilities:

- Conducted user research
- Created personas and user flows
- Designed wireframes and prototypes
- Performed usability testing

  Refined high-fidelity designs

Understanding the user

- User research
- Personas
- Problem statements
- User journey maps

User research: summary

To better understand the needs of parents and students, I conducted user interviews and distributed short surveys focusing on school transportation challenges. Initially, I assumed users mainly needed clearer bus schedules. However, research revealed that safety concerns, real-time updates, and trusted carpool coordination were equally important. Parents prioritized reliability and security, while students valued speed and simplicity. These insights guided the design toward a map-based interface with real-time tracking and a verified carpool group feature.

User research: pain points

1

Unclear Bus Information

Parents and students struggle to find accurate, centralized bus routes and stop details. This guided the design toward a simplified map-based interface with clearly labeled routes and real-time updates.

2

Safety Concerns

Parents are worried about their children's safety during daily commutes. This led to incorporating verified profiles, trusted carpool groups, and notification features into the design.

3

Poor Communication for Carpooling

There is no easy way to coordinate carpool groups within school communities. This influenced the creation of a built-in carpool search and group management feature.

4

Time Pressure & Busy Schedules

Parents and students need quick access to essential information without navigating complex systems. This encouraged a clean, minimal interface with saved routes and one-tap access to key features.

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Persona: Aisha Khan

Problem statement:

Aisha Khan is a 38-year-old working mother of two school-aged children. who needs reliable, centralized bus route information and trusted carpool coordination, because unclear schedules and safety concerns increase her daily stress and reduce her confidence in the commute system.

User characteristics:

- Age: 38
- Education: Master’s Degree (Business Administration)
- Hometown: Kaohsiung, Taiwan
- Family: Married with two children (ages 8 and 12)
- Occupation: Marketing Manager
- Tech-comfortable but time-constrained
- Safety-focused and highly organized

Goals:

- Ensure her children arrive safely and on time
- Access real-time bus updates quickly
- Coordinate trusted carpools with nearby parents
- Reduce morning commute stress

Frustrations:

- Scattered or outdated bus information
- Last-minute schedule changes
- Difficulty finding reliable carpool partners
- Lack of clear communication between parents

Insight:

Working parents value **trust, reliability, and speed** over extra features. Safety transparency and real-time clarity are more important than complex functionality.

Brief story:

Maria works early morning shifts and cannot always accompany her son to the bus stop. She struggles with complicated transport information and worries about safety when she cannot monitor the commute. She needs a straightforward app with clear directions, large readable text, and simple carpool coordination so she can feel confident that her son reaches school safely.

Persona: Daniel Chen

Problem statement:

Daniel Chen is a 17-year-old high school athlete who needs flexible and reliable carpool options because early morning practices and late evening training sessions do not align with regular school bus schedules.

User characteristics:

- Age: 17
- Education: High School (Grade 12)
- Hometown: Taichung, Taiwan
- Family: Lives with parents and older brother
- Occupation: Student & School Basketball Team Member
- Active, disciplined, and time-sensitive
- Comfortable with technology and mobile apps

Goals:

- Arrive on time for early morning sports practice
- Get home safely after late training sessions
- Quickly find or create carpools with teammates
- Reduce dependence on parents for transportation

Frustrations:

- School buses do not operate during early or late hours
- Parents are not always available for drop-offs
- Difficulty coordinating rides manually through group chats
- Risk of being late to practice or important games

Insight:

Student athletes need **flexibility and coordination**, not just fixed schedules. Transportation solutions must adapt to changing practice times and extracurricular activities.

Brief story:

Daniel has basketball practice at 6:00 AM and often finishes late in the evening. The regular school bus does not match his training schedule, and coordinating rides through messaging apps is unreliable. He needs an easy way to organize carpools with teammates who live nearby so he can focus on performance without worrying about transportation.